INDIAN INSTITUTE OF TECHNOLOGY DELHI

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INTERNSHIP REPORT

Arduino Based Embedded Systems

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1 Introduction

This internship focused on the design and implementation of embedded systems using the Arduino UNO platform. The experiments cover basic digital interfacing, sensor integration, display modules, automation, and embedded applications.

The objective of the internship was to strengthen practical knowledge of microcontrollers, embedded programming, hardware interfacing, and real-time control systems through hands-on projects.

The report documents each experiment with its objective, circuit design, implementation methodology, working principle, results, and conclusion.

2 Fundamentals of Arduino Interfacing

2.1 Experiment 0: LED Flash System

2.1.1 Project Name

LED Flash System

2.1.2 Problem Statement

The objective of this experiment is to interface an LED with Arduino UNO and make the LED blink continuously at a fixed interval.

This experiment helps in understanding:

- Digital output pins of Arduino
- Basic Arduino programming structure
- Use of functions in Arduino
- Timing control using `delay()`

2.1.3 Components Used

Table 2.1: Components Used

Component	**Quantity**
Arduino UNO	1
LED	1
100Ω Resistor	1
Breadboard	1
Jumper Wires	Required

2.1.4 Pin Connections

Table 2.2: Pin Connections

Device	Arduino Pin
LED Positive Terminal	D12
LED Negative Terminal	GND

2.1.5 Software Used

- Arduino IDE
- SimulIDE

2.1.6 Circuit Diagram

Experiments-0_flash_flash.png

Figure 2.1: Circuit Diagram of LED Flash System

2.1.7 Source Code

The complete Arduino source code for this experiment is available in the project repository.

Click here to view the Arduino Source Code

2.1.8 Working Principle

1. The LED is connected to digital pin D12 of Arduino UNO.
2. In the `setup()` function, pin D12 is configured as an output pin using `pinMode()`.
3. The `loop()` function continuously calls the `toggle_led()` function.
4. Inside the `toggle_led()` function:
   - The LED turns ON using `digitalWrite(HIGH)`
   - Arduino waits for 1 second using `delay(1000)`
   - The LED turns OFF using `digitalWrite(LOW)`
   - Arduino again waits for 1 second
5. This process repeats continuously, causing the LED to blink repeatedly.

2.1.9 What I Learned

- Arduino UNO digital pins give approximately 5V output.
- Recommended current per digital pin is 20mA.
- Maximum current per digital pin is 40mA.
- Typical LED operating current is around 10mA to 20mA.
- LEDs should be connected using a current limiting resistor.
- Basic LED interfacing with Arduino UNO.

2.1.10 Conclusion

The LED Blinking System using Arduino UNO was successfully implemented and tested.

The LED blinked continuously with a delay of 1 second ON and 1 second OFF. This experiment helped in understanding basic Arduino programming, digital output control, and hardware interfacing.

2.1.11 References

1. Arduino Official Website
   `https://www.arduino.cc/`
2. Arduino UNO Documentation
   `https://docs.arduino.cc/hardware/uno-rev3/`

3. Arduino Tutorials

   https://www.arduino.cc/en/Tutorial/HomePage

4. Arduino IDE Documentation

   https://docs.arduino.cc/software/ide/

5. SimulIDE

   https://simulide.com/

2.2 7-Segment Display Interfacing

Experiment 0 – Go Further 1

2.2.1 Project Name

7-Segment Display Interfacing with Arduino UNO

2.2.2 Objective / Problem Statement

The objective of this project is to interface a common cathode 7-segment display with Arduino UNO and display numbers sequentially.

This experiment helps in understanding:

- 7-segment display interfacing
- Digital output control
- Number display logic
- Multi-pin control using Arduino

2.2.3 Components Used

Table 2.3: Components Used

Component	Quantity
Arduino UNO	1
7-Segment Display (Common Cathode)	1
220Ω Resistors	7
Jumper Wires	Required
Breadboard	1

2.2.4 Pin Connections

Table 2.4: Pin Connections

7-Segment Pin	Arduino Pin
a	D2
b	D3
c	D4
d	D5
e	D6
f	D7
g	D8

2.2.5 Software Used

- Arduino IDE
- SimulIDE

2.2.6 Circuit Diagram

Exp-0/7_segment_led/7_Segment_Led.png

Exp-0/7_segment_led/7-Segment with push botton.png

Figure 2.2: 7-Segment Display Interfacing Circuit

2.2.7 Source Code

The complete Arduino source code for this experiment is available in the project repository.

Click here to view the Arduino Source Code

2.2.8 Working Principle

1. The 7-segment display segments are connected to Arduino digital pins D2 to D8.
2. Each segment is controlled individually using `digitalWrite()`.
3. Depending on which segments are turned ON or OFF, different numbers are displayed.
4. The Arduino sequentially displays:
    - 0
    - 1
    - 2
    - 3
5. Each number remains visible for 1 second using `delay(1000)`.
6. The sequence repeats continuously.

2.2.9 Learning Outcomes

- Interfacing a 7-segment display with Arduino UNO
- Common cathode display working principle
- Using multiple digital output pins
- Number display logic using segments
- Arduino digital output voltage is approximately 5V
- Recommended current per pin is 20mA
- Typical LED segment current is around 10mA to 20mA

2.2.10 Conclusion

The 7-segment display was successfully interfaced with Arduino UNO.

The display sequentially showed numbers from 0 to 3 by controlling individual segments through Arduino digital pins. This project helped in understanding display interfacing and segment control logic.

2.2.11 References

1. Arduino Official Website
   https://www.arduino.cc/
2. Arduino UNO Documentation
   https://docs.arduino.cc/hardware/uno-rev3/
3. 7-Segment Display Basics
   https://www.electronics-tutorials.ws/blog/7-segment-display-tutorial.html
4. Arduino Tutorials
   https://www.arduino.cc/en/Tutorial/HomePage

2.3 LED Matrix Emoji Display System

Experiment 0 (Go Further 2)

2.3.1 Project Name

LED Matrix Emoji Display System

2.3.2 Objective / Problem Statement

The objective of this project is to interface an 8×8 LED matrix with Arduino UNO and display different emoji patterns.

This experiment helps in understanding:

- LED matrix interfacing
- Row and column scanning
- Pattern generation
- Multiplexing technique

2.3.3 Components Used

Table 2.5: Components Used

Component	Quantity
Arduino UNO	1
8×8 LED Matrix	1
125Ω Resistors	8
Jumper Wires	Required
Breadboard	1

2.3.4 Pin Connections

Table 2.6: Pin Connections

LED Matrix	**Arduino Pin**
Row 1	D1
Row 2	D2
Row 3	D3
Row 4	D4
Row 5	D5
Row 6	D6
Row 7	D7
Row 8	D8
Column 1	A5
Column 2	A4
Column 3	A3
Column 4	A2
Column 5	D9
Column 6	D10
Column 7	D11
Column 8	D12

2.3.5 Software Used

- Arduino IDE
- Proteus / SimulIDE

2.3.6 Circuit Diagram

Exp-0/Led_Matrix_8X8/Led_ Matrix_ Circuit.png

Figure 2.3: Circuit Diagram of LED Matrix Emoji Display System

2.3.7 Source Code

The complete Arduino source code for this experiment is available in the project repository.

Click here to view the Arduino Source Code

2.3.8 Working Principle

1. The 8×8 LED matrix is connected to Arduino UNO using row and column connections.
2. Arduino scans rows and columns rapidly using multiplexing.
3. Binary patterns are stored inside the `emoji` array.
4. The `displayEmoji()` function activates rows one by one and controls columns according to the stored pattern.
5. Different emoji patterns are displayed continuously on the LED matrix.
6. Due to fast scanning, the complete image appears continuously visible.

2.3.9 Learning Outcomes

- Interfacing an LED matrix with Arduino UNO
- Multiplexing technique
- Row and column scanning
- Pattern generation using binary values
- Arduino digital output voltage is approximately 5V
- Recommended current per pin is 20mA
- LED matrix LEDs require current limiting resistors

2.3.10 Conclusion

The LED matrix was successfully interfaced with Arduino UNO.

Different emoji patterns were displayed by controlling rows and columns using multiplexing techniques. This project helped in understanding matrix displays and pattern generation in embedded systems.

2.3.11 References

1. Arduino Official Website
   https://www.arduino.cc/
2. Arduino UNO Documentation
   https://docs.arduino.cc/hardware/uno-rev3/
3. LED Matrix Basics
   https://www.electronicsforu.com/resources/learn-electronics/8x8-led-matrix-modul
4. Arduino Tutorials
   https://www.arduino.cc/en/Tutorial/HomePage